scientific reports

OPEN

An enhanced HMM map matching algorithm incorporating personal road selection preferences

Yingxue Zhang[1,2,3], Haowen Yan[1,2,3]✉ & Xiaomin Lu[1,2,3]

Hidden Markov Model (HMM)-based map matching algorithm presents several significant limitations. First, the candidate road segment generation process heavily depends on geometric features while largely ignoring semantic attributes and spatiotemporal context of the road network. Second, the probability modeling phase often fails to account for drivers' individualized road selection preferences. To address these issues, this study proposes an improved HMM-based map matching algorithm that incorporates drivers' personalized road selection preferences (PP-HMM). In the candidate road segment generation stage, a multi-dimensional fused scoring function is constructed by integrating spatial distance, directional similarity, road segment semantic attributes, and temporal factors, enabling more accurate ranking and selection of candidate segments. Moreover, by extending the state transition and observation probabilities of the HMM framework, the proposed method integrates various drivers' personalized road selection preferences observed during actual route selection, including basic route attribute preferences, road network structural characteristics, driving behavior traits, and temporal dynamics, thereby establishing a more comprehensive transition probability model. Experimental comparisons with traditional ST-HMM algorithms demonstrate the enhanced performance and robustness of the proposed approach across diverse road network environments.

Keywords Trajectory, Road network, Road selection, Preference, HMM, Map matching

In recent years, the rapid development of location-based services and mobile social networks has led to the widespread application of Global Positioning System (GPS) technology in vehicles and intelligent terminals. These devices are typically equipped with GPS modules and are capable of collecting large volumes of trajectory data, including mobile location sequences and vehicle status information. GPS-based trajectory data analysis has become a core component in various domains such as traffic navigation, route planning, and driving behavior research. However, due to inherent inaccuracies in GPS devices, complex urban road structures, and signal occlusion, the positioning of GPS-equipped vehicles and mobile entities—represented by GPS sampling points—may deviate from their actual positions on the road network.

Map matching algorithms play a crucial role in intelligent transportation and navigation systems[1,2] by accurately mapping discrete GPS trajectory points onto corresponding road segments (Fig. 1), thereby directly influencing the accuracy and reliability of downstream applications[3,4]. Nevertheless, challenges persist due to uncertainties in GPS signals and the intricate configurations of road networks.

The fundamental objective of map matching is to infer the true position or reconstruct the actual trajectory of a moving object by integrating observed GPS coordinates with the topological or spatial structure of the road network[5]. By determining the precise location of the moving object within the road network based on GPS latitude and longitude data, accurate alignment between GPS sampling points and corresponding road segments can be achieved, enabling effective error correction. Over the past decade, researchers have proposed numerous map matching algorithms to address these challenges. In 2009, Newson and Krumm[6] introduced a HMM-based map matching algorithm that effectively handles the uncertainty of trajectory data and the complexity of road networks, establishing itself as a mainstream approach in this field. This method considers both geometric and topological features, fully exploits trajectory information, and transforms the matching problem into a probabilistic state estimation task, demonstrating strong robustness and adaptability. However, this algorithm exhibits certain limitations: (1) During the candidate road segment generation phase, it primarily relies on geometric features—such as projection distance and direction deviation angle—while neglecting important

[1]Faculty of Geomatics, Lanzhou Jiaotong University, Lanzhou, People's Republic of China. [2]National-Local Joint Engineering Research Center of Technologies and Applications, National Geographic State Monitoring, Lanzhou, People's Republic of China. [3]Gansu Provincial Engineering Laboratory, National Geographic State Monitoring, Lanzhou, People's Republic of China. ✉email: 58391794@qq.com

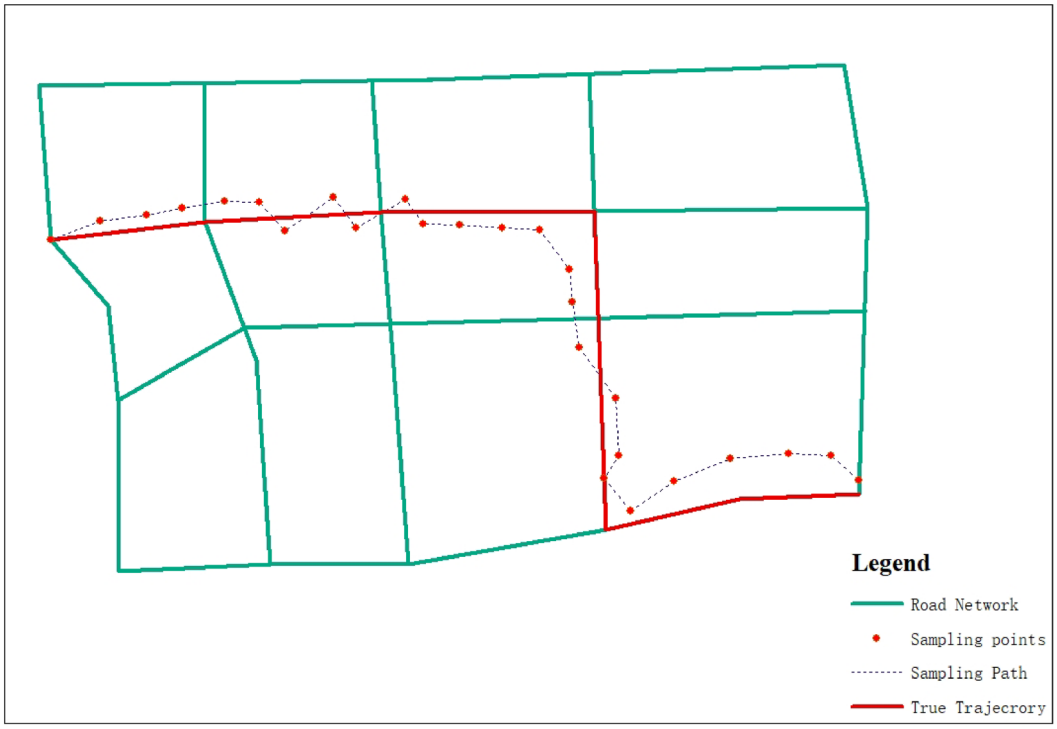


Fig. 1. Schematic diagram of map matching.

semantic attributes of roads (e.g., road level and lane count) and temporal dynamics (e.g., time-dependent variations in road weights); (2) The probabilistic modeling process fails to systematically incorporate drivers' personalized road selection preferences, such as preferences for shortest-time or shortest-distance routes, road grade preferences, and habitual driving behaviors.

To address these issues, this study proposes an enhanced HMM-based map matching algorithm that integrates drivers' personalized road selection preferences. The proposed algorithm improves the accuracy and personalization capabilities of map matching by designing a multi-constraint candidate road segment generation mechanism and extending the HMM probability modeling framework to account for individualized road selection tendencies. Specifically, the main contributions of this work are threefold: (1) A multi-constraint candidate road segment generation mechanism: A composite scoring function incorporating geometric spatial features (spatial distance, directional deviation), road semantic attributes (road grade, lane number), and temporal factors is developed to enable refined scoring and selection of candidate road segments; (2) Comprehensive integration of drivers' personalized road selection preferences: By modifying the state transition and observation probabilities of the HMM, the model incorporates drivers' basic path attribute preferences (shortest time or distance), road network structural feature preferences (average road level, frequency of level changes), driving behavior characteristics (speed habits, frequently traveled paths), and time-varying multi-source preferences; (3) Experimental validation and application: Comparative experiments conducted against the ST-HMM algorithm demonstrate the effectiveness and superior performance of the proposed method across diverse road network environments.

Related work

Classification of map matching algorithms

Extensive research on map matching algorithms has produced substantial advancements, which can be broadly categorized into four primary types: Geometry-based map matching algorithms[7–10], topology-based map matching algorithms[11–14], probability-based map matching algorithms[15–24], and machine learning-based approaches[25–28].

Geometry-based map matching algorithms primarily rely on spatial features such as distance, angle, and shape to perform matching. However, these methods do not consider the connectivity between road segments, resulting in limited stability and accuracy, particularly in complex urban environments. Topology-based algorithms improve upon this by incorporating both proximity and network connectivity from a topological perspective. While they offer enhanced efficiency and accuracy, their performance remains sensitive to data noise and sparse trajectory sampling.

Probability-based map matching algorithms treat GPS coordinates as random variables and trajectories as stochastic processes. These methods define confidence regions—typically elliptical or rectangular—around each sampling point and compute likelihood scores based on spatial distances within those regions. The most probable

path is then determined using probabilistic inference. Despite their theoretical advantages, such methods often suffer from high computational complexity and implementation difficulty.

Advanced algorithms, including Kalman filtering[16], fuzzy logic models[17], and HMM approaches[6], integrate both geometric and topological information. Among them, the HMM algorithm proposed by Newson and Krumm is widely recognized as a classic method in the field, demonstrating strong robustness in handling GPS noise and complex road networks. Nevertheless, traditional HMM algorithms predominantly depend on geometric and topological features, neglecting drivers' personalized road selection preferences—an important factor in real-world applications.

With the advancement of machine learning technologies, researchers have increasingly applied deep learning and reinforcement learning techniques to map matching tasks. For example, some studies leverage trajectory data to train neural network models for predicting the corresponding roads of trajectory points. However, these methods typically require large-scale labeled datasets and face challenges related to model interpretability, limiting their practical deployment.

Although the aforementioned methods have achieved promising results in various scenarios, they still exhibit limitations in addressing complex traffic environments and individual driving behaviors. Notably, few studies have explored how drivers' personalized road selection preferences influence map matching outcomes, despite its significance in real-world navigation systems. Therefore, integrating personalized road selection preferences into map matching algorithms represents a critical direction for future research.

Optimization of HMM map matching algorithm accuracy

Improvements in the accuracy of classical HMM-based map matching algorithms can be grouped into three major categories: (1) Enhancing the probabilistic modeling framework to refine accuracy. For instance, Liu et al.[29] introduced driving direction into transition probability calculations. The ST-Matching algorithm[30] integrates spatiotemporal features, geometric topology, and speed constraints, employing Gaussian distributions to model observation probabilities and computing transition probabilities based on vehicle speed, GPS observations, and measured positions. (2) Pre- and post-processing of trajectory data to enhance matching precision. Zhou[31] proposed manually labeling error-prone points before applying standard HMM matching to other points. Wu et al.[32] used genetic algorithms to further optimize HMM-derived matching results, improving segment-level accuracy. Wöltche[33] developed a novel approach combining Markov decision processes and reinforcement learning to dynamically handle outliers and noise clusters. (3) Incorporating additional data sources to improve accuracy. For example, the EHMM algorithm[34] and STD-Matching algorithm[35] utilize instantaneous directional information, the OHMM algorithm[36] incorporates road width data, and the IVMM algorithm[37] introduces speed-related parameters to enhance matching precision.

Optimization of map matching algorithm efficiency

Efficiency optimization is a crucial requirement in practical implementations of map matching algorithms: (1) Reducing computational overhead through controlled expansion of candidate road segments. Huang et al.[38] proposed extracting common patterns from historical trajectory data to limit the search space of candidate segments, thereby improving efficiency. Raymond et al.[39] replaced road segments with intersections, transforming line-point analysis into point-point analysis and significantly reducing geometric computation complexity. Shiteng[40] and Xie[41] improved efficiency by segmenting trajectory data and minimizing the number of state transition probability computations. (2) Utilizing spatial indexing structures to accelerate candidate road segment retrieval. Song et al.[42] implemented R-tree indexing to build an efficient road network index, enabling rapid identification of sub-trajectory-associated road segments. This method further enhances efficiency by analyzing position change rates to segment trajectories and focusing on key points rather than entire paths. Xiaomeng et al.[43] proposed a grid-based map matching model that uses hash indexing for fast candidate segment lookup, effectively lowering time complexity. Wei et al.[44] and Yuguang[45] constructed indexing structures based on grid partitioning and road network rasterization, enabling rapid localization of candidate segments via binary search algorithms.

Current limitations of map matching algorithms

Despite significant progress in map matching technology, existing algorithms still face challenges in handling complex traffic conditions and capturing individual driving behaviors. In particular, few studies have considered the impact of drivers' personalized road selection preferences on matching accuracy, although this factor plays a vital role in real-world applications. For example, Ceikute and Jensen[46] proposed an improved navigation service by comparing driver behavior with trajectory data in terms of route length, popularity, and travel time. Dai et al.[47] leveraged large-scale trajectory data to provide personalized driving recommendations. Zheng et al.[48] developed a system that infers users' historical travel paths by analyzing behavioral patterns and integrating them into route reasoning. Huang et al.[38] identified recurring patterns in historical trajectory data and applied them to map matching tasks. Gao et al.[49] proposed a preference-driven matching algorithm that combines local driving habits with global dynamic programming for road network alignment. Moreover, current algorithms often struggle to balance computational efficiency and accuracy when dealing with high-density, high-dynamic traffic data. Real-time performance and scalability are especially critical in large-scale deployments, where algorithms must deliver accurate results rapidly. Hence, developing efficient and reliable map matching algorithms capable of adapting to complex environments and personalized needs remains a key research focus.

In conclusion, integrating drivers' personalized road selection preferences into map matching algorithms while enhancing real-time performance and accuracy constitutes one of the most pressing challenges in future research.

Problem definition

Terminology and notes

Several formal definitions related to map matching in this paper are as follows:

Definition 1 A road network is formally represented by a directed graph $G = \{V, E\}$, where V denotes the set of all vertices and road intersections, E represents the set of one-way road segments within the road network. Each vertex v_i can be mathematically expressed as $v_i = \{id, lon, lat\}$, and each road segment e_i can be expressed as $e_i = \{id, v_{start}, v_{end}, length\}$.

Definition 2 A path P in a road network is a set of road segments $P = e_1 \rightarrow e_2 \rightarrow \cdots \rightarrow e_n$, where e_1, e_n are the origin and end of a path.

Definition 3 A trajectory T is defined as a sequence of sampling points obtained from the same moving object in chronological order, denoted as $T_i = \{t_1, t_2 \cdots t_n\}$, where n represents the number of sampling points contained in a segment of the trajectory. Each trajectory point t_i can be expressed as $t_i = \{id_i, time_i, lon_i, lat_i\}$.

Definition 4 Candidate road segments refer to a set of potential road segments identified based on predefined rules within the vicinity of a target trajectory point (GPS sampling points, GPS observation points). Candidate points are the corresponding matching points located on these candidate road segments that align with the target trajectory points.

Definition 5 The problem of map matching can be defined as, given a floating car trajectory T and a road network $G = \{V, E\}$, find the path P from $G = \{V, E\}$ that matches T with its real path.

HMM-based map matching

HMM is a probabilistic framework widely employed in time series analysis. It describes a process where a hidden Markov chain generates an unobservable state sequence, and each state stochastically produces an observable value sequence[38]. In the context of a road network, it is assumed that the movement of a moving object follows a Markov process, meaning the current state depends solely on its preceding state and is independent of earlier historical states. While the actual road trajectory traversed by the moving object cannot be directly observed, its corresponding outputs can be inferred from GPS sampling points. Leveraging this principle, the HMM model can estimate the most probable hidden state sequence that corresponds to the observed sequence, thereby reconstructing the actual motion trajectory based on a series of GPS sampling points.

In HMM-based map matching algorithms, the location information collected by GPS devices is treated as the set of observations, while the actual positions of these devices are modeled as the set of hidden states. This transformation allows the map matching problem to be reformulated as a state estimation problem within the framework of HMM. The HMM model is characterized by three key components: The initial state probability matrix π, the state transition probability matrix A, and the observation likelihood probability matrix B:

$$\lambda = (\pi, A, B) \tag{1}$$

State transition probability

The state transition probability quantifies the likelihood that a moving object will transition from one candidate road segment e_i to another e_j, which based on the following two assumptions: First, the closer the distance between two actual trajectory points, the higher the probability of state transition. Second, the smaller the discrepancy between the distance of two real trajectory points on the road segment and the distance of two GPS-observed sample points, the greater the probability of state transition. Consequently, the state transition probability $P(e_j \,|e_i\,)$ is calculated by the following formula[2]:

$$P(e_j \,|e_i\,) = \frac{1}{\beta} \exp\left(-\frac{d_t}{\beta}\right) \tag{2}$$

where d_t denotes the difference between the distance of two real trajectory points on the road segment or the distance of two GPS-observed sample points, and β represents the tolerance parameter for non-straight arrivals.

Observation probability

The observation probability quantifies the degree to which a GPS sampling point o_t corresponds to its surrounding candidate road segments e_j. Typically, the observation probability is inversely proportional to the distance between the GPS sample point and the candidate road segment. In other words, the closer the sample point is to the candidate road segment, the higher the likelihood that its true location lies on the road segment. Based on the Gaussian distribution assumption, the observation probability $P(o_t \,|e_j\,)$ can be calculated using the following formula[2]:

$$P\left(o_t \,|s_j\,\right) = \frac{1}{\sqrt{2\pi} \cdot \sigma} \exp\left(\frac{d}{2\sigma^2}\right) \tag{3}$$

where σ represents the GPS positioning error and d denotes the perpendicular distance from the trajectory point o_t to the candidate road segment e_j.

Enhanced HMM map matching algorithm incorporating personal road selection preferences

System framework

The proposed algorithmic framework, as illustrated in Fig. 2, consists of two primary components: Candidate Road Set Generation and Feature Analysis. During the data preprocessing phase, trajectory data and road network data are systematically processed to generate the candidate road set. In the Candidate Road Set Generation stage, a multi-dimensional fusion scoring approach is applied to accurately rank and filter the candidate segments, thereby identifying the most probable matching road segments. The Feature Analysis phase focuses on refining the observation and transition probabilities: The observation probability integrates both spatial distance and

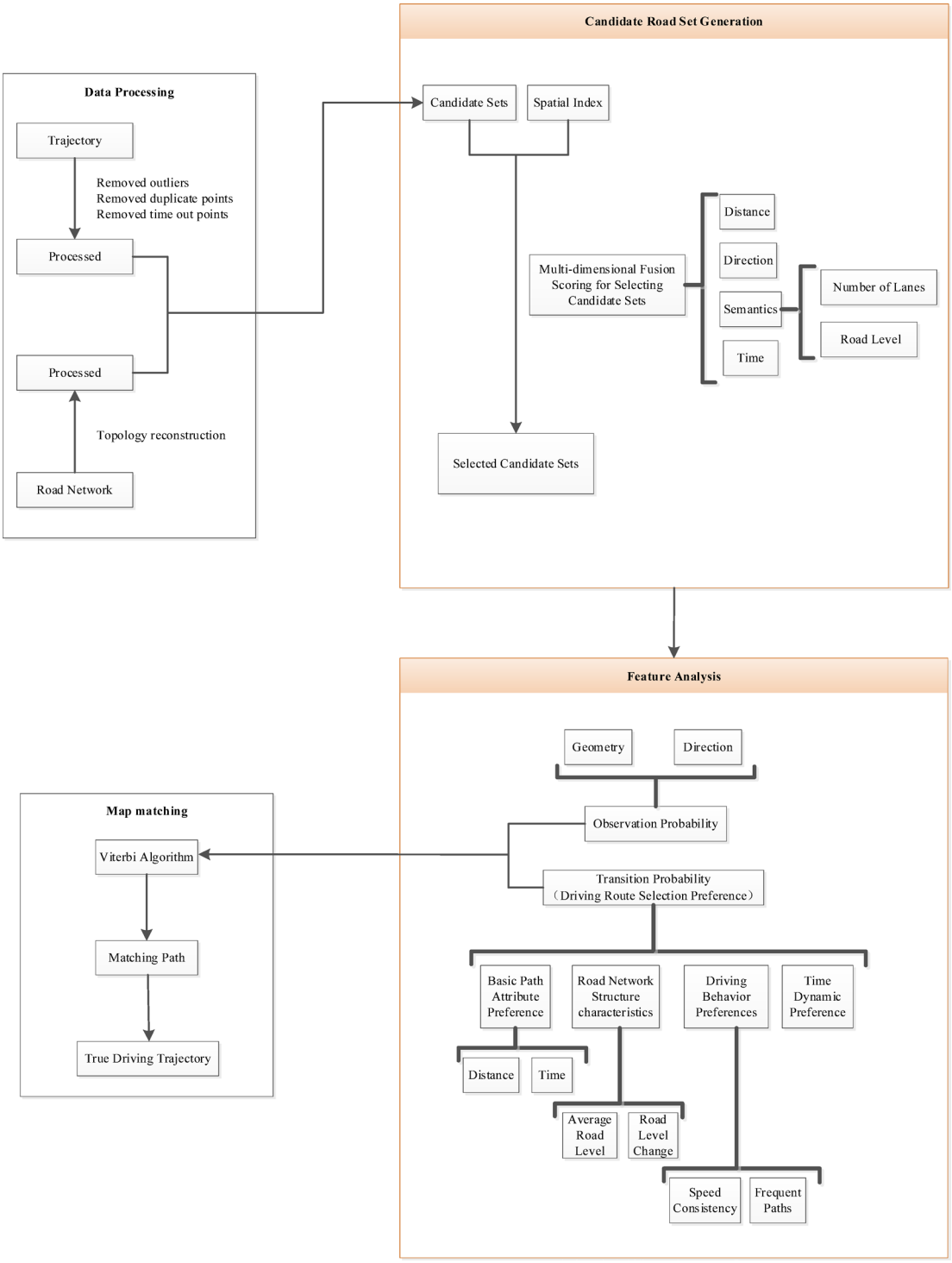


Fig. 2. Framework of proposed algorithm.

directional deviation, while the transition probability incorporates the influence of drivers' personalized road selection preferences. Finally, the Viterbi algorithm is employed to determine the most likely actual trajectory.

Candidate road segment generation

In the HMM-based map matching process, the generation of candidate road segments represents a critical stage in model construction. The primary objective of this phase is to identify potential road segments from the entire road network that are associated with the current trajectory point, thereby forming the basis for subsequent calculations of state transition probabilities and observation probabilities. In this study, the candidate road segment generation process begins with the application of an adaptive R-tree-based spatial indexing technique, which efficiently narrows down the search space and reduces computational overhead across the entire road network. Subsequently, a multi-dimensional fusion scoring mechanism is applied to accurately rank and filter the candidate set, enabling the selection of the most probable matching road segments.

Adaptive R-tree spatial indexing

This study applies an adaptive R-tree-based spatial indexing method to generate candidate road segment sets. To account for the spatial distribution characteristics of GPS positioning errors, the retrieval radius R is dynamically determined based on error ellipses:

$$R = \max(a, b) + 3\sigma \tag{4}$$

where a and b denote the major and minor axes of the error ellipse, respectively, and σ represents the standard deviation of historical positioning errors.

Multi-dimensional fusion scoring

In the candidate road segment generation phase, this research introduces a comprehensive multi-dimensional fusion scoring function that integrates spatial distance, directional deviation, road segment semantic attributes, and temporal factors to evaluate and prioritize candidate road segments. The multi-dimensional fusion scoring function is formulated as:

$$score(e_i) = \exp\left(-\frac{d}{\sigma_d}\right) \cdot \exp\left(-\frac{\Delta\theta}{\sigma_\theta}\right) \cdot \omega_{semantic}(e_i) \cdot \omega_{time}(t) \tag{5}$$

where $\exp\left(-\frac{d}{\sigma_d}\right)$ and $\exp\left(-\frac{\Delta\theta}{\sigma_\theta}\right)$ represent the spatial distance and directional consistency between the trajectory point and candidate road segment e_i, respectively, $\omega_{semantic}(e_i)$ denotes the semantic attributes of road segment e_i, including its road level and lane count, $\omega_{time}(t)$ reflects the temporal influence, capturing dynamic variations in road conditions across different time periods.

The semantic attribute $\omega_{semantic}(e_i)$ is expressed as:

$$\omega_{semantic}(e_i) = \alpha \cdot \omega_{level}(e_i) + (1 - \alpha) \cdot \omega_{lane}(e_i) \tag{6}$$

where $\omega_{level}(e_i)$ indicates the road classification weight, with values of 1, 0.8, 0.6, 0.4, 0.3, and 0.2 assigned to motorway, trunk, primary, secondary, tertiary, and residential classifications, respectively, $\omega_{lane}(e_i)$ represents the lane count correction coefficient, defined as:

$$\omega_{lane}(e_i) = \min\left(1, \frac{n_{lane}(e_i)}{n_{\max}}\right)^{\gamma} \tag{7}$$

where $n_{lane}(e_i)$ denotes the number of lanes on road segment e_i, $n_{\max}$ represents the maximum lane count within the road network, and γ serves as an empirical adjustment factor.

The temporal attribute $\omega_{time}(t)$ is defined as:

$$\omega_{time}(t) = \sum_{k=1} \beta_k \cdot \exp\left(-\frac{(t - \mu_k)^2}{2\sigma_k^2}\right) \tag{8}$$

where μ_k denotes the central time of time period k, σ_k represents the standard deviation of the time interval, and β_k indicates the base weight assigned to time period k.

By integrating these components, the proposed methodology enables a more accurate assessment of the matching potential of candidate road segments, thereby significantly improving both the precision and computational efficiency of map matching algorithms.

Drivers' personalized road selection preferences

Drivers' personalized road selection preferences refers to the differential choice behavior exhibited by individuals based on their utility evaluation of available routes within a multi-modal transportation environment[50]. Research suggests that experienced drivers engage in a complex risk–benefit trade-off during route decision-making, where the criteria extend beyond minimizing expected travel time to include assessments of delay risk probability and potential for travel time optimization[51,52]. Specifically, drivers with an optimistic risk attitude tend to choose the theoretically shortest-time path while underestimating the impact of traffic congestion[53], whereas risk-

averse drivers place significant emphasis on route familiarity, aligning with the principles of cognitive economy theory[54]. From a probabilistic standpoint, when a road segment exhibits low overall cost and high individual preference probability, its likelihood of being included in the actual trajectory increases significantly[55]. This study identifies seven core route selection factors grounded in transportation behavior theory and the discrete choice model framework, which are integrated into the HMM map matching algorithm.

These core route selection factors include:

Basic route attributes Minimum travel distance[56] and minimum travel time[57] serve as fundamental variables in conventional route selection decisions;

Road network structural characteristics Average road grade[58] and frequency of road grade changes[59], reflecting driving comfort and cognitive workload;

Driving behavior preferences Speed habits[60] represent individual tendencies in speed selection, while habitual routes[61] indicate path dependency behaviors;

Temporal dynamic factors Time-of-day patterns[62] capture variations in commuter decision-making during peak and off-peak periods.

Enhancement of transition probabilities

In conventional HMM, the computation of transition probabilities primarily depends on geometric distances between road segments or trajectory points. However, this approach has inherent limitations, as it fails to fully capture the multi-dimensional decision-making preferences that drivers exhibit during real-world route selection. To improve the model's ability to characterize the rationality of continuous trajectories, this study incorporates seven core route selection factors: Minimum travel distance, minimum travel time, average road grade, frequency of road grade changes, speed preferences, habitual route usage, and temporal patterns. These factors collectively form the basis for constructing a more comprehensive and behavior-aware transition probability function.

Traditional transition probability models rely exclusively on geometric distances, thereby overlooking the influence of drivers' personalized road selection behaviors. To address this limitation, the present study extends the classical transition probability framework into a joint probabilistic model that integrates seven categories of drivers' personalized road selection preferences. This enhancement enables a more accurate representation of dynamic transition tendencies and strengthens the modeling of trajectory continuity and rationality. The improved transition probability function is formulated as:

$$P(e_j \,|e_i\,) = \frac{1}{Z}\prod f_k^{\beta_k} \tag{9}$$

where Z denotes the normalization factor ensuring that the total probability sums to 1, β_k represents the elasticity coefficient associated with the k-th route selection preference feature function, and f_k corresponds to the respective feature function, with detailed definitions provided in the following sections.

Fundamental path attribute preferences

The fundamental path attribute preferences include the preference for shortest travel time f_{time} and the preference for shortest travel distance f_{dis}:

$$f_{time} = \exp\left(-\frac{t_{ij} - t_{\min}}{t_{median}}\right) \tag{10}$$

$$f_{dis} = \exp\left(-\frac{d_{ij} - d_{\min}}{d_{avg}}\right) \tag{11}$$

where t_{ij} and d_{ij} represent the estimated travel time and geometric distance from candidate segment e_i to segment e_j, respectively. $t_{\min}$ and $d_{\min}$ denote the minimum travel time and shortest distance among all reachable candidate segments, t_{median} represents the median travel time across all current candidate segments, and d_{avg} signifies the average travel distance of all current candidate segments.

Road network structural feature preferences

The road network structural feature preferences consist of the average road level preference f_{level} and the road level change:

$$f_{level} = \frac{r_{e_i}}{r_{\max}} \tag{12}$$

$$f_{change} = \exp\left(-\lambda_{change} \cdot \left|r_{e_i} - r_{e_j}\right|\right) \tag{13}$$

where r_{e_i} indicates the standardized road grade score of candidate segment e_i, and λ_{change} represents the penalty coefficient associated with road level change, when transitioning from candidate segment e_i to e_j.

Driving behavior preferences

Driving behavior preferences include speed consistency preference f_{speed} and frequently path preference f_{freq}:

$$f_{speed} = \mathrm{N}(v_{e_i}\,|\mu_v, \sigma_v^2\,) \tag{14}$$

$$f_{freq} = \varepsilon + \frac{count(e_i \to e_j)}{\max(count(e_i \to *))} \tag{15}$$

where v_{e_i} denotes the posted speed limit or real-time observed speed on candidate segment e_i, μ_v represents the driver's historical average speed on similar roads, and σ_v indicates the variance of historical speeds, reflecting the stability of the driver's behavior. $count(e_i \to e_j)$ is the number of transitions from segment e_i to e_j observed in historical trajectories, $count(e_i \to *)$ is the total number of transitions originating from segment e_i, and ε serves as a smoothing coefficient to prevent zero probabilities.

Temporal dynamic factor preferences
The temporal dynamic factor preference function f_{time} captures periodic driving behavior patterns through time-dependent weight adjustments:

$$f_{time} = \alpha_{peak} \cdot I_{peak}(t) + \alpha_{night} \cdot I_{night}(t) + \alpha_{off} \cdot I_{off}(t) \tag{16}$$

where t denotes the current time, I_{peak}, I_{night} and I_{off} are indicator functions representing morning/evening peak hours, nighttime, and other time periods, respectively, α_{peak}, α_{off} and α_{night} are corresponding preference weights assigned to each period, and $\alpha_{peak} + \alpha_{night} + \alpha_{off} = 1$ ensures normalized weighting.

Enhancement of observation probability
Within the framework of HMM-based map matching, optimizing the observation probability plays a critical role in reducing localization uncertainty and improving road segment matching accuracy. Traditional approaches to observation probability modeling primarily rely on geometric distance while neglecting directional deviation. To address this limitation, the refined observation probability model integrates both the perpendicular projection distance from trajectory points to candidate road segments and the angular deviation between vehicle heading and road direction. This enhanced model is formulated using a Gaussian distribution as follows:

$$P(o_t \,|e_j\,) = \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot \exp\left(-\frac{d_\perp^2}{2\sigma_d^2} - \frac{\Delta\theta^2}{2\sigma_\theta^2}\right) \tag{17}$$

where $d_\perp$ denotes the perpendicular projection distance from trajectory point $o_t o_t$ to candidate road segment e_j, $\Delta\theta$ represents the angle between the vehicle's heading direction and the orientation of road segment, and σ_d and σ_θ are parameters characterizing the noise distributions associated with distance and directional measurements, respectively.

Algorithmic solution
The optimal matching path problem within the HMM framework can be effectively solved using the Viterbi algorithm, a dynamic programming technique. The Viterbi algorithm identifies the most probable sequence of hidden states—corresponding to the optimal sequence of matched road segments—by iteratively computing joint probabilities based on observation and transition probabilities. At each step, it records the preceding state that leads to the current maximum likelihood, ultimately determining the endpoint of the optimal path.

For the enhanced HMM map matching algorithm that incorporates drivers' personalized road selection preferences, these behavioral factors—including shortest travel distance, minimal travel time, average road grade, frequency of road grade changes, speed habits, habitual route usage, and temporal patterns—require holistic optimization across the entire trajectory rather than isolated point-level matching. The Viterbi algorithm ensures that the resulting path satisfies both geometric consistency (local spatial constraints) and behavioral continuity by recursively computing cumulative probabilities. The recursive formula for the maximum likelihood probability is defined as:

$$\delta_t(e_j) = \max\left[\delta_{t-1}(e_i) \cdot P(e_j \,|e_i\,)\right] \cdot P(o_t \,|e_j\,) \tag{18}$$

where $\delta_t(e_j)$ denotes the maximum likelihood probability of road segment e_j at time t, and $P(e_j \,|e_i\,)^*$ and $P(o_t \,|e_j\,)$ represent the enhanced transition probability and observation probability, respectively. Through this recursive computation, the final optimal matching path is determined, and the complete path is reconstructed by backtracking through the recorded predecessor states.

After completing road segment-level matching, point-level matching is further performed. The procedure is as follows: If the target trajectory point can be directly projected onto the optimal matching road segment, the projection point is selected as the matching location, if the projection falls outside the segment (e.g., on its extension), the distances between the trajectory point and both endpoints of the road segment are computed, and the closer endpoint is chosen as the matching point.

By integrating these enhancements, the proposed HMM-based map matching algorithm not only achieves accurate road segment-level matching but also enables precise localization of trajectory points on the road network, thereby significantly improving both the accuracy and robustness of the overall map matching process.

Experiment and discussion
Experimental data and settings
In this study, the GPS trajectory data of taxis or online ride-hailing services from representative Chinese cities (Beijing, Hangzhou, and Wuxi) were selected as the test dataset. The raw data included taxi ID, UTC timestamps,

Outlier processing	Preprocessing
Out of limit point	BeiJing[116.17,116.62,39.83,40.05]
	HangZhou[118.21,120.30,29.11,30.33]
	WuXi[119.33,120.38,31.7,32.2]
Point of saturation	Five consecutive data points exhibit a moving distance of less than 10 m and a speed of less than 1 m per second
Point of drift	Instantaneous speed exceeds 120 km/h or acceleration exceeds 3.6 m/s^2
Point of repetition	The time interval between adjacent points is less than 1 s, and the displacement is less than 0.5 m

Table 1. Preprocessing of trajectory data.

Regions	Number of regional trajectories	Number of trajectory points	Number of road network nodes	Number of road network edges
BeiJing	500	112,287	32,781	22,796
HangZhou	5000	1,223,374	80,943	43,967
WuXi	150	204,983	46,850	26,726

Table 2. Study region.

Regions	Algorithm	Average accuracy (%)	Efficiency of single-point matching(ms)
BeiJing	ST-HMM	86.37	5770
	PP-HMM	94.90	238
HangZhou	ST-HMM	78.29	5982
	PP-HMM	94.11	251
WuXi	ST-HMM	76.63	5631
	PP-HMM	94.41	169

Table 3. Comparison of algorithm performance.

and latitude/longitude coordinates, etc. Following the preprocessing of Table 1, the subsequent outliers were excluded:

The road network data were constructed using OpenStreetMap, and the topological relationships were validated through ArcGIS 10.8. The processed trajectory and road network data are presented in Table 2.

Evaluation indicators

In this study, the accuracy and average accuracy of map matching are employed as evaluation metrics to assess the quality of the matching model. The calculation formulas are presented as follows:

$$AC_i = \frac{m_i}{n_i} \tag{19}$$

$$\overline{AC_i} = \frac{\sum_{i=1}^{i=T} \frac{m_i}{n_i}}{T} \tag{20}$$

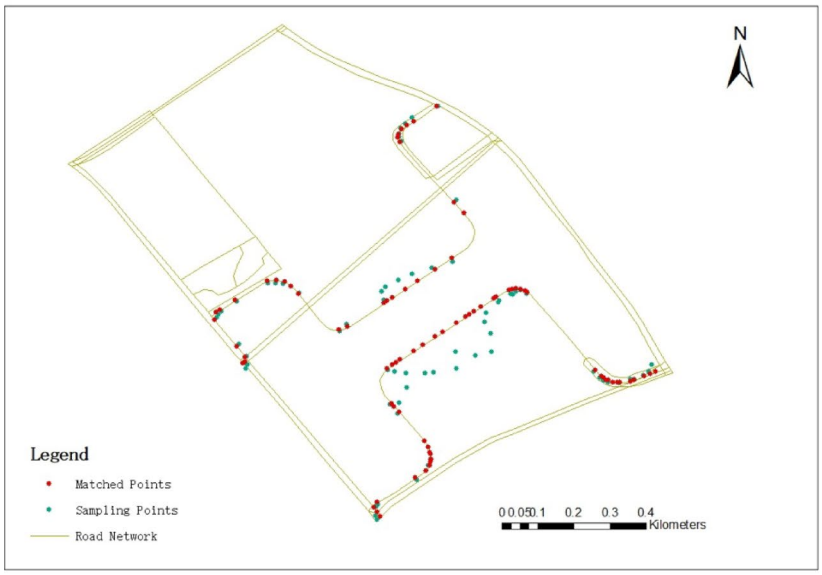
where, AC_i represents the accuracy of the ith trajectory, where m_i denotes the number of trajectory points correctly matched to the corresponding road segment in the ith trajectory, and n represents the total number of all trajectory points to be matched in the ith trajectory. Additionally, $\overline{AC_i}$ represent the average accuracy, and T denote the total number of trajectories.

Analysis of Results

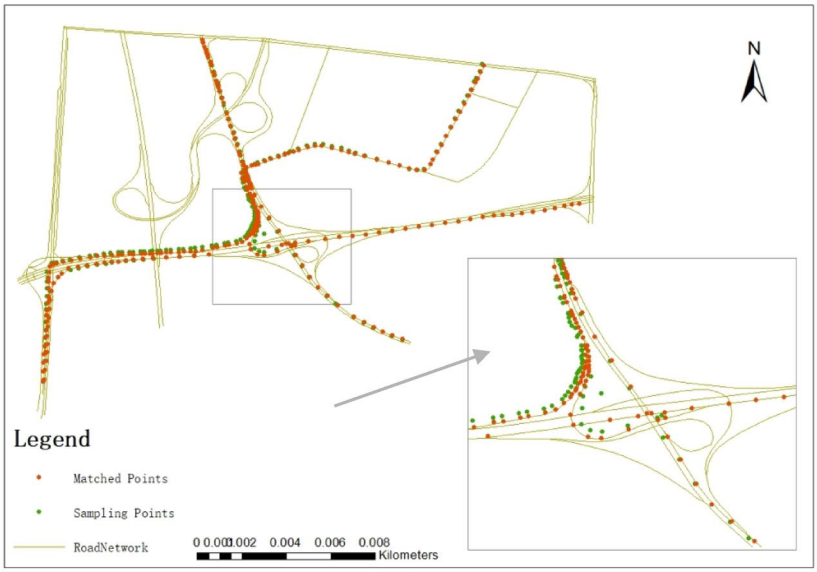
Comparative analysis of matching accuracy

As presented in Table 3, the comparative performance analysis clearly indicates that the PP-HMM algorithm significantly outperforms the classical ST-HMM algorithm across road networks in three distinct cities.

In the Beijing region, when matching 500 trajectories, the PP-HMM algorithm achieved a minimum trajectory accuracy of 85.10%. This particular trajectory consisted of 141 trajectory points, with 120 successfully matched, and 93 trajectories were fully matched with 100% accuracy. In Wuxi, among 150 matched trajectories, the PP-HMM algorithm attained a minimum trajectory accuracy of 79.35%, involving a trajectory with 982 points, of which 661 were correctly matched, and 39 trajectories achieved perfect matching. In Hangzhou, during the evaluation of 5000 trajectories, the PP-HMM algorithm recorded a minimum trajectory accuracy of 72.43%, with one representative trajectory containing 555 points, of which 402 were accurately matched, and 1788 trajectories achieved complete matching.



(a) Map matching results for ordinary road sections



(b) Map matching results for complex road sections

Fig. 3. Example of map matching result.

Compared to the ST-HMM algorithm, the PP-HMM algorithm demonstrates considerable improvements in average precision. Under the road network conditions of Beijing, Hangzhou, and Wuxi, the PP-HMM algorithm achieved average precision rates of 94.90%, 94.11%, and 94.41%, respectively—substantially surpassing those of the ST-HMM algorithm. Moreover, the PP-HMM algorithm exhibits a significant reduction in single-point processing time, with average durations of 238 ms, 251 ms, and 169 ms observed in Beijing, Hangzhou, and Wuxi, respectively, indicating a notable enhancement in computational efficiency.

Analysis of typical cases

From the trajectory comparison diagram before and after matching (Fig. 3), it is evident that the distribution of trajectory points prior to matching is relatively dispersed and disorganized, making it challenging to accurately reflect the actual road conditions of the vehicle. In contrast, the matched trajectory points closely align with the edges of the road network, enabling precise reconstruction of the vehicle's driving path. In complex road sections such as intersections and turns, unmatched trajectory points are prone to deviation from the actual road, whereas the matched trajectory remains well-correlated with the real road in these areas. This clearly demonstrates the

robust adaptability of the PP-HMM algorithm in complex road scenarios, allowing for accurate identification of vehicle trajectories under intricate road conditions. Furthermore, pre-matching trajectories may exhibit jitter due to GPS signal errors. After matching, this jitter is effectively mitigated, resulting in a more stable trajectory across the entire driving path. This further substantiates the significant efficacy of the PP-HMM algorithm in enhancing both the accuracy and stability of trajectory data.

Conclusion

This study addresses the limitations of conventional HMM-based map-matching algorithms by proposing an enhanced HMM algorithm—PP-HMM, which incorporates drivers' individualized road selection preferences. Through the construction of a multi-dimensional fusion scoring function, the algorithm enables more precise filtering of candidate road segments by integrating geometric spatial features, road semantic attributes, and spatiotemporal contextual information. Furthermore, during the probability modeling phase, the algorithm systematically incorporates multiple aspects of driver behavior, including drivers' basic path attribute preferences (shortest time or distance), road network structural feature preferences (average road level, frequency of level changes), driving behavior characteristics (speed habits, frequently traveled paths), and time-varying multi-source preferences. This integration enhances the algorithm's alignment with real-world driving patterns.

Experimental results demonstrate that the PP-HMM algorithm significantly outperforms the ST-HMM algorithm in terms of matching accuracy across diverse urban road networks. Specifically, the PP-HMM achieved average precision rates of 94.90%, 94.11%, and 94.41% in the road networks of Beijing, Hangzhou, and Wuxi, respectively, along with a notable reduction in single-point processing time. In-depth case studies further confirm the algorithm's effectiveness in handling complex road environments and improving trajectory stability.

In summary, the PP-HMM algorithm offers a more efficient, accurate, and personalized approach to map matching, making a meaningful contribution to the development of intelligent transportation systems. Future research directions include incorporating additional real-time traffic data, such as traffic flow and congestion levels, which enhance the algorithm's adaptability in dynamic environments. Moreover, optimizing computational complexity and exploring strategies for efficient deployment on resource-constrained platforms, such as in-vehicle navigation systems and mobile devices, that can broaden its practical applicability and enable more accurate location-based services for end users.

Data availability

Data are available in a publicly accessible repository that does not issue DOIs. Publicly available datasets were analyzed in this study. This data can be found here: https://www.microsoft.com/en-us/research/publication/t-drive-trajectory-data-sample/.

Received: 18 October 2024; Accepted: 29 July 2025
Published online: 15 October 2025

7. References

1. Zhou, C. et al. Survey of map-matching algorithm for intelligent trans-port system. *Comput. Sci.* **42**(10), 1–6 (2015).
2. Teng, Z. et al. Vehicle navigation map matching algorithm based on multiple-attribute integration strategy. *J. Jiangsu Univ. (Nat. Sci. Ed.)* **39**(114–118), 25 (2018).
3. Zhao, Y. et al. A new method of road network updating based on floating car data. In *2011 IEEE International Geoscience and Remote Sensing Symposium* 1878–1881 (2011).
4. Ge, Y. et al. A taxi driving fraud detection system. In *2011 IEEE 11th International Conference on Data Mining* 181–190 (2011).
5. Yu, J., Yang, Q., Lu, J., Han, J. & Peng, H. Advanced map-matching algorithms: A survey and trends. *Acta Electron. Sin.* **49**, 1818–1829 (2021).
6. Newson, P. & Krumm, J. Hidden Markov map matching through noise and sparseness. In *Proceedings of the 17th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems* 336–343 (2009).
7. Bang, Y., Kim, J. & Yu, K. An improved map-matching technique based on the Fréchet distance approach for pedestrian navigation services. *Sensors* **16**, 142–8220 (2016).
8. Taylor, G., Blewitt, G., Steup, D., Corbett, S. & Car, A. Road reduction filtering for GPS-GIS navigation. *Trans. GIS* **5**, 193–207 (2001).
9. An, L. et al. Map matching algorithm based on course vertical line. *Sci. Surv. Mapp.* **42**(06), 112–116 (2017).
10. Teng, W. & Wang, Y. Real-time map matching: A new algorithm integrating spatio-temporal proximity and improved weighted circle. *Sci. Lett.* **11**, 288–297 (2019).
11. Yuan, L., Li, D. & Hu, S. A map-matching algorithm with low-frequency floating car data based on matching Path. *EURASIP J. Wirel. Commun. Netw.* **2018**, 1687–1499 (2018).
12. Liu, Y. Forecast map-matching model for vehicle-borne navigation based on roadway characteristic. In *2010 International Conference on Optoelectronics and Image Processing* (IEEE, 2010).
13. Zheng, S. et al. A map matching algorithm based on discrete Frechet distance. *J. Hefei Univ. Technol. (Nat. Sci.)* **40**(01), 42–46 (2017).
14. Zeng, Z. et al. Curvature integration constrained map matching method for GPS floating car data. *Acta Geod. Cartogr. Sin.* **44**(10), 1167–1176 (2015).
15. Ren, M. & Karimi, H. A. A hidden Markov model-based map-matching algorithm for wheelchair navigation. *J. Navig.* **62**, 383–395 (2009).
16. Xu, H., Liu, H., Tan, C. W. & Bao, Y. Development and application of an enhanced Kalman filter and global positioning system error-correction approach for improved map-matching. *J. Intell. Transp. Syst.* **14**, 27–36 (2010).
17. Zhang, T., Yang, D., Li, K. & Lian, X. Fuzzy map-matching algorithm with confidence feedback for vehicle navigation. *J. Tsinghua Univ. (Sci. Technol.)* **49**, 277–280 (2009).
18. Choi, S. T., Hur, W. S. & Seo, S. W. Cooperative localization based on topology matching. In *Proceedings of the 2014 IEEE 6th International Symposium on Wireless Vehicular Communications* 1–5 (2014).
19. Tuo, Y., Yun, X. & Zhang, Y. MSTMA novel map matching approach for low- sampling- rate trajectories. In *Proceedings of the 2017 IEEE 28th Annual International Symposium on Personal, Indoor, and Mobile Radio Communications (PIMRC)* 1–7 (2017).

20. Li, X. et al. Map matching algorithm design based on GPS information. In *Proceedings of the 2014 IEEE Workshop on Electronics, Computer and Applications* 898–901 (2014).
21. Song, Y. et al. enhanced HMM map matching model based on multiple type trajectories. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining* (Springer, 2024).
22. Sun, M. & Li, X. Traffic trajectory data analysis technology based on HMM model map matching algorithm. *PLoS ONE* **19**, 0302656 (2024).
23. Qu, L., Zhou, Y., Li, J., Yu, Q. & Jiang, X. HMM-based map matching and spatio temporal analysis for matching errors with taxi trajectories. *ISPRS Int. J. Geo Inf.* **12**, 330 (2023).
24. Singh, S., Singh, J., Goyal, S. B., El Barachi, M. & Kumar, M. Analytical review of map matching algorithms: Analyzing the performance and efficiency using road dataset of the Indian subcontinent. *Arch. Computat. Methods Eng.* **30**, 4897–4916 (2023).
25. Jiang, L., Chen, C. X. & Chen, C. L2mm: Learning to map matching with deep models for low-quality GPS trajectory data. *ACM Trans. Knowl. Discov. Data* **17**(3), 1–25 (2023).
26. Zhang, Z. et al. Learning visual semantic map-matching for loosely multi-sensor fusion localization of autonomous vehicles. *IEEE Trans. Intell. Veh.* **8**(1), 358–367 (2022).
27. Shen, Z. et al. Dmm: A deep reinforcement learning based map matching framework for cellular data. *IEEE Trans. Knowl. Data Eng.* **36**, 5120–5137 (2024).
28. Jin, Z. et al. Transformer-based map-matching model with limited labeled data using transfer-learning approach. *Trans. Res. Part C Emerg. Technol.* **140**, 103668 (2022).
29. Liu, W., Li, M., Xu, X. & Mao, S. An improved map-matching algorithm based on HMM Model. *Acta Sci. Nat. Univ. Pekin.* **54**, 1235–1241 (2018).
30. Lou, Y. et al. Map-matching for low-sampling-rate GPS trajectories. *ACM SIGSPATIAL GIS. Ac. Med.* 352–361 (2009).
31. Xibo, Z., Ding, Y., Haoyu, T., Luo, Q. & Ni, L. M. HIMM: An HMM-based interactive map-matching system. In *22nd International Conference on Database Systems for Advanced Applications* 3–18 (Springer, 2017). https://doi.org/10.1007/978-3-319-55699-4_1.
32. Wu, G., Qiu, Y. J. & Wang, G. R. Map-matching algorithm based on hidden Markov model and genetic algorithm. *J. Northeast. Univ. (Nat. Sci.)* **38**, 472–475 (2017).
33. Wöltche, A. Open source map matching with Markov decision processes: A new method and a detailed benchmark with existing approaches. *Trans. GIS* **27**(7), 1959–1991 (2023).
34. Che, M. L., Wang, Y. L., Zhang, C. & Cao, X. An enhanced hidden Markov map matching model for floating car data. *Sensors* **18**, 1758 (2018).
35. Hsueh, Y. L. & Chen, H. C. Map matching for low-sampling-rate GPS trajectories by exploring real-time moving directions. *Inf. Sci.* **433–434**, 55–69 (2018).
36. Goh, C. Y. et al. Online map-matching based on hidden Markov model for real-time traffic sensing applications. In *Proceedings of the 2012 15th International IEEE Conference on Intelligent Transportation Systems* (IEEE, 2012).
37. Yuan, J., Zheng, Y., Zhang, C., Xie, X. & Sun, G. Z. An interactive-voting based map matching algorithm. In *Eleventh International Conference on Mobile Data Management* 43–52 (IEEE, 2010).
38. Huang, Y. et al. Frequent pattern-based map-matching on low sampling rate trajectories. In *19th IEEE International Conference on Mobile Data Management* (IEEE, 2018).
39. Raymond, R., Morimura, T. & Osogami, T. Map-matching with hidden Markov model on sampled road network. *Pattern Recognit. (ICPR)* 2242–2245.
40. Shiteng, W. *Research on Map-Matching Method Based on Track Segmentation and Improved HMM Model* (Jiangxi Univ. of Science and Technology, 2023).
41. Xie, Y., Zhou, K., Miao, F. & Zhang, Q. High-accuracy off-line map-matching of trajectory network division based on weight adaptation HMM. *IEEE Access* **8**, 7256–7266 (2020).
42. Song, Y. et al. Efficient HMM map-matching method using R-tree and trajectory segmentation. *J. Syst. Simul.* **35**, 339–349 (2023).
43. Xiaomeng, W., Tianhe, C. & Hui, L. A research of map-matching method for massive floating car data. *Geogr. Inf. Sci.* **17**, 1143–1151 (2015).
44. Wei, Z., Jian-Min, X. U. & Mian-feng, L. I. N. Map-matching algorithm of large scale probe vehicle data. *J. Transp. Syst. Eng. Inf. Technol.* **7**, 39–45 (2007).
45. Yuguang, L. & Qingquan, L. Fast road matching algorithm of massive floating car databased on map rasterization. *Geom. Inf. Sci. Wuhan Univ.* **39**(724–728), 733 (2014).
46. Ceikute, V. & Jensen, C. S. Local driver behavior versus routing services. In *14th International Conference on Mobile Data Management* (IEEE, 2013).
47. Dai, J., Yang, B., Guo, C. & Ding, Z. Personalized route recommendation using Big Trajectory Data. In *31st International Conference on Data Engineering* (IEEE, 2015).
48. Zheng, K., Zheng, Y., Xie, X. & Zhou, X. Reducing uncertainty of low-sampling-rate trajectories. In *28th International Conference on Data Engineering* (IEEE, 2012).
49. Gao, X., Wu, Y. J., Guo, L. M., Ding, Z. M. & Chen, J. C. Personalized map-matching algorithm based on driving preference. *RuanJianXueBao J. Softw.* **29**, 3500–3516 (2018).
50. Ben-Akiva, M. E. & Lerman, S. R. *Discrete Choice Analysis: Theory and Application to Travel Demand* (MIT press, 1985).
51. Avineri, E. & Prashker, J. N. The impact of travel time information on travelers' learning under uncertainty. *Transportation* **3**(4), 393–408. https://doi.org/10.1007/s11116-005-5710-y (2006).
52. Mahmassani, H. S. & Liu, Y. H. Dynamics of commuting decision behaviour under Advanced Traveller Information Systems. *Trans. Res. Part C Emerg. Technol.* **7**(2/3), 91–107. https://doi.org/10.1016/S0968-090X(99)00014-5 (1999).
53. Dia, H. & Panwai, S. Modelling drivers' compliance and route choice behaviour in response to travel information. *Nonlinear Dyn.* **49**, 493–509 (2007).
54. Gärling, T. & Golledge, R. G. Cognitive mapping and spatial decision-making. In *Cognitive Mapping* 44–65 (Routledge, 2018).
55. Prato, C. G. Route choice modeling: Past, present and future research directions. *J. Choice Modell.* **2**(1), 65–100 (2009).
56. Dijkstra, E. W. A note on two problems in connexion with graphs. *Numer. Math.* https://doi.org/10.1007/BF01386390 (1959).
57. Xing, T. & Zhou, X. Finding the most reliable path with and without link travel time correlation: A Lagrangian substitution based approach. *Trans. Res. Part B Methodol.* **45**(10), 1660–1679 (2011).
58. Ni, J. et al. The impacts of the traffic situation, road conditions, and driving environment on driver stress: A systematic review. *Transp. Res. Part F Psychol. Behav.* **103**, 141–162. https://doi.org/10.1016/j.trf.2024.04.006 (2024).
59. Oron-Gilad, T. & Ronen, A. Road characteristics and driver fatigue: A simulator study. *Traffic Injury Prev.* **8**(3), 281–289. https://doi.org/10.1080/15389580701354318 (2007).
60. Taubman-Ben-Ari, O., Mikulincer, M. & Gillath, O. The multidimensional driving style inventory–scale construct and validation. *Accid. Anal. Prev.* **36**(3), 323–332. https://doi.org/10.1016/S0001-4575(03)00010-1 (2004).
61. Puffert, D. J. Path dependence. In *Handbook of Cliometrics* 1–28 (Springer, 2023).
62. Kant, P., Machavarapu, P. K. & Kiran, S. Modelling of mode choice behaviour during off-peak hours—A case of New Delhi City. In *International Conference on Transportation System Engineering and Management* 347–371 (Springer, 2023).

Acknowledgements
This study was funded by the National Natural Science Foundation of China (Grant No. 41930101); Regional Fund of National Natural Science Foundation of China (Grant No. 42161066); and the Industrial Support and Program Project of Universities in Gansu Province (Grant No. 2022CYZC-30).

Author contributions
Conceptualization, Y.Z.; methodology, Y.Z.; software, Y.Z.; validation, Y.Z.; formal analysis, Y.Z.; investigation, Y.Z.; resources, H.Y. and X.L.; data curation, Y.Z.; writing—original draft preparation, Y.Z.; writing—review and editing, Y.Z.; visualization, Y.Z.; supervision, H.Y. and X.L.; project administration, X.L.; funding acquisition, H.Y. and X.L. All authors have read and agreed to the published version of the manuscript.

Funding
This study was funded by the National Natural Science Foundation of China (Grant No. 41930101); Regional Fund of National Natural Science Foundation of China (Grant No.42161066); and the Industrial Support and Program Project of Universities in Gansu Province (Grant No.2022CYZC-30).

Declarations

Competing interests
The authors declare no competing interests.

Additional information
Correspondence and requests for materials should be addressed to H.Y.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2025